POWER + PiBot INTERFACE (taps SPI / limits / rails)

+3V3
+5V
GND
C1
10uF
C2
10uF
J1
SPI_SCK
SPI_MOSI
SPI_MISO
MCP_CS
MCP_INT
TO_PIBOT
D2
LED_A
PWR
R1
1k
HX_SCK
HX_DOUT
J2
TO_PIBOT_BC

CAN-FD: MCP2518FD + TJA1051 + split termination + TVS

C3
100nF
U1
MCP2518FD
SPI_MOSI
SPI_MISO
MCP_CS
SPI_SCK
MCP_INT
MCP_RXCAN
MCP_TXCAN
OSC1
OSC2
Y1
40MHz
C5
18pF
C4
18pF
U2
TJA1051T-3
C6
100nF
C7
4.7nF
CANH
CANL
D1
NUP2105L
RT1
60R
RT2
60R
CAN_T
J3
CAN

LOAD-CELL FRONT-END: HX711 (5V exc / 3V3 I/O)

U3
HX711
HX_BASE
HX_DOUT
HX_SCK
AVDD
VFB
R2
20k
R3
8.2k
C8
10uF
C9
100nF
C12
100nF
C10
100nF
C11
10uF
R4
1k
R5
1k
C13
100nF
LC_A_P
LC_A_N
J4
LOADCELL

Microspark
Sheet: /
File: MS-EDM-IFACE.kicad_sch
Title: MS-EDM-IFACE PiBot EDM Interface Adapter (CAN-FD + load cell
Size: A3
Date: 2026-06-16
Rev: A
KiCad E.D.A. 10.0.3
Id: 1/1